Online Resource 1 — Seen or new? Duplicate records, conformal coverage on novel inputs, and a novelty-conditional remedy for tabular benchmarks

All tables are generated by `supplementary/build_supplement.py` from the result files in `results/`. Coverage gaps are in proportions (multiply by 100 for percentage points).

S1 Dataset audit

One row per dataset (OpenML data ID). `n_raw`: rows before group-preserving subsampling to 30,000. `dup_row_frac`: share of rows in a group of identical feature vectors of size ≥ 2. `conflict_row_frac`: share of rows in groups with more than one label. `pred_exposure`: Proposition 2. `obs_exposure`: mean over 10 random splits. `set`: DUP ($\pi \geq 0.01$), LOW ($0 < \pi < 0.01$), CLEAN (no duplicates).

task	data_id	name	n_rows	n_raw	n_features	n_classes	dup_row_frac	conflict_row_frac	max_group	pred_exposure	obs_exposure	set
classification	469	analcatdata_dmft	797	797	4	6.000	0.893	0.862	29	0.811	0.810	DUP
classification	4534	PhishingWebsites	11055	11055	30	2.000	0.713	0.032	25	0.576	0.572	DUP
classification	1067	kc1	2109	2109	21	2.000	0.519	0.120	112	0.475	0.478	DUP
classification	15	breast-w	699	699	9	2.000	0.406	0.000	27	0.377	0.379	DUP
classification	1464	blood-transfusion-service-center	748	748	4	2.000	0.421	0.267	35	0.367	0.374	DUP
classification	1063	kc2	522	522	21	2.000	0.356	0.226	63	0.320	0.299	DUP
classification	40978	Internet-Advertisements	3279	3279	1558	2.000	0.347	0.016	58	0.300	0.291	DUP
classification	1053	jm1	10885	10885	21	2.000	0.251	0.082	173	0.214	0.215	DUP
classification	1068	pc1	1109	1109	21	2.000	0.226	0.012	31	0.171	0.167	DUP
classification	40984	segment	2310	2310	18	7.000	0.203	0.002	6	0.126	0.122	DUP
classification	1475	first-order-theorem-proving	6118	6118	51	6.000	0.173	0.055	21	0.121	0.123	DUP
classification	44	spambase	4601	4601	57	2.000	0.125	0.001	69	0.099	0.096	DUP
classification	23	cmc	1473	1473	9	3.000	0.143	0.095	4	0.095	0.089	DUP
classification	1050	pc3	1563	1563	37	2.000	0.123	0.006	35	0.094	0.104	DUP
classification	1049	pc4	1458	1458	37	2.000	0.115	0.002	31	0.091	0.096	DUP
classification	6	letter	20000	20000	16	26.000	0.109	0.000	26	0.081	0.078	DUP
classification	1486	nomao	30000	34465	118	2.000	0.106	0.000	136	0.079	0.079	DUP
classification	40670	dna	3186	3186	180	3.000	0.097	0.001	5	0.071	0.069	DUP
classification	46	splice	3190	3190	60	3.000	0.097	0.001	5	0.071	0.072	DUP
classification	18	mfeat-morphological	2000	2000	6	10.000	0.111	0.107	4	0.067	0.066	DUP
classification	1468	cnae-9	1080	1080	856	9.000	0.047	0.000	4	0.032	0.035	DUP
classification	1480	ilpd	583	583	10	2.000	0.045	0.000	2	0.027	0.024	DUP
classification	1462	banknote-authentication	1372	1372	4	2.000	0.026	0.000	4	0.022	0.019	DUP
classification	38	sick	3772	3772	29	2.000	0.025	0.000	6	0.020	0.021	DUP
classification	40499	texture	5500	5500	40	11.000	0.010	0.000	2	0.006	0.007	LOW
classification	40983	wilt	4839	4839	5	2.000	0.008	0.000	2	0.005	0.004	LOW
classification	14	mfeat-fourier	2000	2000	76	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	12	mfeat-factors	2000	2000	216	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	22	mfeat-zernike	2000	2000	47	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	40979	mfeat-pixel	2000	2000	240	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	16	mfeat-karhunen	2000	2000	64	10.000	0.006	0.000	2	0.004	0.003	LOW
classification	1487	ozone-level-8hr	2534	2534	72	2.000	0.004	0.000	9	0.004	0.003	LOW
classification	1494	qsar-biodeg	1055	1055	41	2.000	0.006	0.000	2	0.003	0.001	LOW
classification	23381	dresses-sales	500	500	12	2.000	0.004	0.004	2	0.002	0.004	LOW
classification	1489	phoneme	5404	5404	5	2.000	0.003	0.000	2	0.002	0.002	LOW
classification	1590	adult	30000	48842	14	2.000	0.002	0.000	2	0.001	0.001	LOW
classification	4538	GesturePhaseSegmentationProcessed	9873	9873	32	5.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	188	eucalyptus	736	736	19	5.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40994	climate-model-simulation-crashes	540	540	18	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	151	electricity	30000	45312	8	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	458	analcatdata_authorship	841	841	70	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	37	diabetes	768	768	8	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	54	vehicle	846	846	18	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	182	satimage	6430	6430	36	6.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	50	tic-tac-toe	958	958	9	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1510	wdbc	569	569	30	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	3	kr-vs-kp	3196	3196	36	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	31	credit-g	1000	1000	20	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1501	semeion	1593	1593	256	10.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	11	balance-scale	625	625	4	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1485	madelon	2600	2600	500	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	307	vowel	990	990	12	11.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40975	car	1728	1728	6	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	4134	Bioresponse	3751	3751	1776	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1461	bank-marketing	30000	45211	16	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40966	MiceProtein	1080	1080	77	8.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	28	optdigits	5620	5620	64	10.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1497	wall-robot-navigation	5456	5456	24	4.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	1478	har	10299	10299	561	6.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40668	connect-4	30000	67557	42	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	41027	jungle_chess_2pcs_raw_endgame_complete	30000	44819	6	3.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	300	isolet	7797	7797	617	26.000	0.000	0.000	1	0.000	0.000	CLEAN

task	data_id	name	n_rows	n_raw	n_features	n_classes	dup_row_frac	conflict_row_frac	max_group	pred_exposure	obs_exposure	set
classification	40701	churn	5000	5000	20	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	6332	cylinder-bands	540	540	37	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	29	credit-approval	690	690	15	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	40982	steel-plates-fault	1941	1941	27	7.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	23517	numerai28.6	30000	96320	21	2.000	0.000	0.000	1	0.000	0.000	CLEAN
classification	32	pendigits	10992	10992	16	10.000	0.000	0.000	1	0.000	0.000	CLEAN
regression	44984	cps88wages	28155	28155	6		0.898	0.898	71	0.853	0.853	DUP
regression	44966	solar_flare	1066	1066	10		0.894	0.665	73	0.848	0.854	DUP
regression	44964	superconductivity	21263	21263	81		0.400	0.396	110	0.334	0.334	DUP
regression	44971	white_wine	4898	4898	11		0.349	0.000	8	0.232	0.234	DUP
regression	44972	red_wine	1599	1599	11		0.288	0.000	4	0.181	0.181	DUP
regression	44970	QSAR_fish_toxicity	908	908	6		0.239	0.239	7	0.180	0.186	DUP
regression	44976	sarcos	30000	48933	21		0.182	0.000	2	0.109	0.109	DUP
regression	44993	health_insurance	22272	22272	11		0.124	0.088	19	0.096	0.097	DUP
regression	44990	brazilian_houses	10692	10692	9		0.107	0.058	24	0.081	0.083	DUP
regression	44963	physiochemical_protein	30000	45730	9		0.073	0.002	9	0.048	0.047	DUP
regression	44959	concrete_compressive_strength	1030	1030	8		0.051	0.019	4	0.041	0.040	DUP
regression	44962	forest_fires	517	517	12		0.048	0.033	3	0.030	0.030	DUP
regression	44979	diamonds	30000	53940	9		0.013	0.007	5	0.008	0.008	LOW
regression	44994	cars	804	804	17		0.002	0.002	2	0.001	0.002	LOW
regression	44974	video_transcoding	30000	68784	18		0.002	0.002	2	0.001	0.001	LOW
regression	44983	miami_housing	13932	13932	15		0.001	0.001	3	0.001	0.001	LOW
regression	44989	kings_county	21613	21613	21		0.000	0.000	2	0.000	0.000	LOW
regression	44987	socmob	1156	1156	5		0.000	0.000	1	0.000	0.000	CLEAN
regression	44992	fps_benchmark	24624	24624	43		0.000	0.000	1	0.000	0.000	CLEAN
regression	44973	grid_stability	10000	10000	12		0.000	0.000	1	0.000	0.000	CLEAN
regression	44965	geographical_origin_of_music	1059	1059	116		0.000	0.000	1	0.000	0.000	CLEAN
regression	44958	auction_verification	2043	2043	7		0.000	0.000	1	0.000	0.000	CLEAN
regression	45402	space_ga	3107	3107	6		0.000	0.000	1	0.000	0.000	CLEAN
regression	44957	airfoil_self_noise	1503	1503	5		0.000	0.000	1	0.000	0.000	CLEAN
regression	44969	naval_propulsion_plant	11934	11934	14		0.000	0.000	1	0.000	0.000	CLEAN
regression	44960	energy_efficiency	768	768	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	44980	kin8nm	8192	8192	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	44977	california_housing	20640	20640	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	45012	fifa	19178	19178	28		0.000	0.000	1	0.000	0.000	CLEAN
regression	44956	abalone	4177	4177	8		0.000	0.000	1	0.000	0.000	CLEAN
regression	41021	Moneyball	1232	1232	14		0.000	0.000	1	0.000	0.000	CLEAN
regression	44981	pumadyn32nh	8192	8192	32		0.000	0.000	1	0.000	0.000	CLEAN
regression	44978	cpu_activity	8192	8192	21		0.000	0.000	1	0.000	0.000	CLEAN
regression	44975	wave_energy	30000	72000	48		0.000	0.000	1	0.000	0.000	CLEAN
regression	44967	student_performance_por	649	649	30		0.000	0.000	1	0.000	0.000	CLEAN

S2 Protocol deviations and post-hoc analyses

Freeze commit: `0e58bdd`. Everything below happened **after** the freeze and is disclosed in the manuscript.

#	When	What	Why	Effect on confirmatory results
1	Pipeline debugging, after the freeze but before full runs	`run_injection` / `run_capacity` now create their output directories (a bug fix found in quick mode)	Crash in quick mode	None
2	After the main results (primary H1 **not** supported at the median level)	Added **exploratory** analyses: (a) exposure strata π < 0.1, 0.1–0.3, ≥ 0.3; (b) the Proposition 1 decomposition of each dataset × learner gap into a seen-over-coverage term and a marginal term; (c) counts of dataset × learner pairs with gaps below −0.02 and −0.05; (d) Fig. 8	To explain *where* the gap arises. The theory says exposure alone is not sufficient: seen rows must also be over-covered. These analyses test that consequence	None. All pre-registered tests are reported unchanged and the new analyses are labelled exploratory
3	After the main results | **Verification run** on solar_flare (OpenML 44966): 200 extra random splits (seeds 1000–1199), ridge and LightGBM, absolute-residual score, α = 0.1 (`code/experiments/verify_solar_flare.py`)	With 10 seeds, MCP novel coverage on this dataset was 0.84–0.88, apparently contradicting Proposition 3	Confirms that it was sampling noise: MCP novel coverage 0.909 (95% CI 0.899–0.918) for LightGBM and 0.907 (0.896–0.917) for ridge; SCP novel coverage 0.78 and 0.77. Per-split SD ≈ 0.07 with about 36 novel calibration rows. Reported as a verification and a stated limitation (high split-to-split variability when novel calibration sets are small)	
4	After review round 1 (Reviews 1 and 5: suites are curated, and CTR23 excludes grouped data), **before any outcome on these data** | **External validation set**, specified in advance: every NASA MDP defect dataset on OpenML not in CC18 with ≥ 300 rows (mc1 #1056, pc2 #1069, kc3 #1065, mw1 #1071), plus KDDCup99 (10% version, #1113; documented heavy redundancy, Tavallaee et al. 2009). Same configuration as the main grid (10 seeds, 5 learners, both protocols). KDDCup99 (494,020 rows) is **uniformly row-subsampled** to 30,000 rows (seed 2026), because group-preserving subsampling cannot keep groups larger than the cap. Pre-specified test: novel-record gap of SCP-R per dataset and learner (LAC, α = 0.1), with seed-level 95% t-intervals; and the MCP-R novel gap	Tests whether the mechanism appears where duplication is known to be common	Additional evidence only; the confirmatory suite analysis is unchanged	
5	Review round 1 (Review 7)	Experiment G (computational cost of the audit and of MCP vs SCP), pre-registered but not yet run, is run now	Omission	None
6	Review round 2 (re-targeting to Pattern Recognition; anticipated baseline request), **before outcomes** | **Normalized-residual baseline** (Papadopoulos et al. 2011): regression DUP datasets (12), RF and LightGBM, 10 seeds, α = 0.1. Score $\lvert y - \hat{y} \rvert / (\hat{\sigma}(x) + \beta)$, where $\hat{\sigma}(x)$ is the mean distance to the k = 10 nearest training rows in the standardized feature space and β is the median of $\hat{\sigma}$ over calibration rows. Compared as SCP-R and MCP-R with the same score. Pre-specified question: does the adaptive score alone remove the novel-record gap?	A locally adaptive score shrinks intervals near training copies and might absorb the effect	Additional baseline; confirmatory results unchanged	

No dataset, seed or learner was excluded after outcomes were seen. No hyperparameter was changed.

S3a Main results on DUP datasets, α = 0.05

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.951	0.974	0.948	-0.002	-0.027	1.000	-0.551	0.005	0.952	0.952	-0.001	0.026	1.007	0.992	1.010
classification	logreg	aps	24	0.953	0.962	0.951	0.001	-0.011	1.000	-0.527	0.008	0.953	0.953	0.000	0.049	1.003	1.002	1.002
classification	rf	lac	24	0.952	0.990	0.947	-0.003	-0.026	1.000	-0.403	0.051	0.953	0.952	0.000	0.032	1.007	0.980	1.007

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	rf	aps	24	0.952	0.949	0.954	0.004	-0.011	1.000	-0.049	0.821	0.954	0.952	-0.000	0.996	0.993	1.107	1.006
classification	lgbm	lac	24	0.951	0.983	0.946	-0.004	-0.040	0.019	-0.598	0.002	0.952	0.951	-0.001	0.000	1.006	0.984	1.007
classification	lgbm	aps	24	0.952	0.966	0.949	-0.001	-0.032	1.000	-0.707	0.000	0.952	0.952	-0.001	0.004	1.006	1.000	1.005
classification	knn	lac	24	0.973	0.995	0.970	0.020	-0.026	1.000	-0.328	0.118	0.975	0.974	0.020	0.031	1.000	0.982	1.004
classification	knn	aps	24	0.957	0.980	0.954	0.004	-0.010	1.000	-0.118	0.582	0.959	0.957	0.002	0.022	1.002	0.999	1.001
classification	mlp	lac	24	0.950	0.977	0.947	-0.003	-0.029	1.000	-0.437	0.033	0.951	0.951	-0.001	0.022	1.009	0.997	1.007
classification	mlp	aps	24	0.952	0.965	0.950	0.000	-0.027	1.000	-0.409	0.047	0.953	0.952	0.000	0.026	1.005	0.999	1.002
regression	logreg	absres	12	0.949	0.960	0.948	-0.002	-0.106	1.000	-0.587	0.045	0.950	0.950	-0.004	0.077	1.017	0.948	1.005
regression	rf	absres	12	0.951	0.989	0.946	-0.004	-0.093	1.000	-0.636	0.026	0.952	0.952	-0.000	0.032	1.039	0.655	0.990
regression	lgbm	absres	12	0.950	0.986	0.946	-0.004	-0.102	1.000	-0.650	0.022	0.952	0.952	-0.001	0.041	1.043	0.789	0.999
regression	knn	absres	12	0.950	0.978	0.948	-0.002	-0.082	1.000	-0.636	0.026	0.952	0.952	-0.001	0.068	1.042	0.833	1.001
regression	mlp	absres	12	0.950	0.977	0.946	-0.004	-0.073	1.000	-0.622	0.031	0.950	0.950	-0.000	0.068	1.032	0.894	1.001

S3b Main results on DUP datasets, α = 0.1

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.901	0.937	0.895	-0.005	-0.065	0.343	-0.583	0.003	0.903	0.902	-0.000	0.001	1.009	0.962	1.008
classification	logreg	aps	24	0.900	0.913	0.899	-0.001	-0.015	1.000	-0.637	0.001	0.903	0.902	-0.001	0.107	1.005	0.997	1.004
classification	rf	lac	24	0.901	0.974	0.894	-0.006	-0.065	0.420	-0.414	0.044	0.903	0.901	-0.001	0.107	1.012	0.950	1.006
classification	rf	aps	24	0.903	0.891	0.905	0.005	-0.012	1.000	-0.118	0.582	0.903	0.901	-0.001	0.979	0.995	1.054	1.001
classification	lgbm	lac	24	0.900	0.968	0.894	-0.006	-0.067	0.092	-0.461	0.023	0.901	0.900	-0.002	0.005	1.011	0.928	1.004
classification	lgbm	aps	24	0.901	0.920	0.898	-0.002	-0.042	0.607	-0.838	0.000	0.901	0.902	-0.002	0.002	1.003	0.988	1.001
classification	knn	lac	24	0.934	0.978	0.926	0.026	-0.028	1.000	-0.356	0.088	0.938	0.936	0.026	0.061	1.000	0.975	1.009
classification	knn	aps	24	0.902	0.919	0.901	0.001	-0.016	1.000	-0.114	0.595	0.903	0.902	-0.001	0.229	1.001	0.992	1.001
classification	mlp	lac	24	0.900	0.947	0.896	-0.004	-0.069	0.395	-0.337	0.107	0.901	0.902	-0.001	0.001	1.012	0.939	1.005
classification	mlp	aps	24	0.901	0.909	0.900	0.000	-0.049	1.000	-0.628	0.001	0.903	0.903	0.000	0.002	1.005	0.990	1.005
regression	logreg	absres	12	0.901	0.920	0.898	-0.002	-0.168	1.000	-0.804	0.002	0.901	0.900	-0.005	0.154	1.026	0.938	1.001
regression	rf	absres	12	0.901	0.976	0.892	-0.008	-0.158	0.881	-0.685	0.014	0.900	0.900	-0.002	0.107	1.043	0.514	0.987
regression	lgbm	absres	12	0.900	0.965	0.894	-0.006	-0.169	1.000	-0.685	0.014	0.900	0.900	-0.001	0.061	1.027	0.782	0.994
regression	knn	absres	12	0.902	0.956	0.898	-0.002	-0.117	1.000	-0.371	0.236	0.902	0.900	-0.003	0.107	1.044	0.820	0.999
regression	mlp	absres	12	0.899	0.941	0.895	-0.005	-0.108	0.833	-0.622	0.031	0.900	0.899	-0.004	0.154	1.027	0.844	1.000
regression	cqr	cqr	12	0.903	0.965	0.894	-0.006	-0.037	0.881	-0.552	0.063	0.903	0.900	-0.001	0.011	1.017	0.954	1.006

S3c Main results on DUP datasets, α = 0.2

Medians over datasets of seed-averaged metrics; split CP with a random split (scp) and novelty-Mondrian CP (mcp).

task	model	score	n_datasets	scp_all_cov_median	scp_seen_cov_median	scp_novel_cov_median	scp_gap_median	scp_gap_min	p_H1_holm	spearman_exposure_gap	p_spearman	mcp_all_cov_median	mcp_novel_cov_median	mcp_gap_ci_lo	p_H2_holm	size_ratio_novel_median	size_ratio_seen_median	size_ratio_all_median
classification	logreg	lac	24	0.799	0.841	0.794	-0.006	-0.165	0.220	-0.326	0.120	0.803	0.803	-0.001	0.003	1.010	0.921	1.003
classification	logreg	aps	24	0.802	0.811	0.799	-0.001	-0.029	1.000	-0.516	0.010	0.803	0.801	-0.002	0.045	1.003	0.993	1.002
classification	rf	lac	24	0.801	0.941	0.787	-0.013	-0.164	0.297	-0.486	0.016	0.804	0.802	-0.003	0.036	1.020	0.882	1.004
classification	rf	aps	24	0.802	0.783	0.806	0.006	-0.021	1.000	-0.274	0.195	0.803	0.804	0.000	0.984	0.997	1.038	1.001
classification	lgbm	lac	24	0.801	0.932	0.789	-0.011	-0.150	0.044	-0.410	0.047	0.800	0.802	-0.003	0.006	1.019	0.869	1.002
classification	lgbm	aps	24	0.801	0.827	0.799	-0.001	-0.081	0.678	-0.677	0.000	0.802	0.803	-0.000	0.007	1.006	0.984	1.001
classification	knn	lac	24	0.849	0.942	0.839	0.039	-0.068	1.000	-0.449	0.028	0.864	0.858	0.044	0.039	1.000	0.966	1.000
classification	knn	aps	24	0.802	0.822	0.800	0.000	-0.034	1.000	-0.274	0.195	0.803	0.803	-0.002	0.045	1.007	0.993	1.002
classification	mlp	lac	24	0.801	0.862	0.792	-0.008	-0.154	0.363	-0.294	0.163	0.803	0.803	-0.001	0.003	1.013	0.895	1.007
classification	mlp	aps	24	0.801	0.818	0.798	-0.002	-0.051	1.000	-0.397	0.054	0.801	0.801	-0.001	0.045	1.003	0.991	1.002
regression	logreg	absres	12	0.802	0.825	0.797	-0.003	-0.258	1.000	-0.671	0.017	0.802	0.801	-0.005	0.151	1.005	0.955	1.001
regression	rf	absres	12	0.799	0.961	0.784	-0.016	-0.247	0.424	-0.566	0.055	0.799	0.799	-0.003	0.042	1.060	0.640	1.000
regression	lgbm	absres	12	0.800	0.912	0.790	-0.010	-0.253	1.000	-0.615	0.033	0.800	0.800	-0.003	0.045	1.037	0.774	1.000
regression	knn	absres	12	0.801	0.878	0.794	-0.006	-0.204	1.000	-0.594	0.042	0.802	0.801	-0.001	0.128	1.037	0.800	1.008
regression	mlp	absres	12	0.801	0.852	0.791	-0.009	-0.212	1.000	-0.636	0.026	0.801	0.800	-0.003	0.128	1.030	0.854	1.001

S4 Group-aware split (SCP-G), α = 0.1

task	model	score	n	g_cov_median	gap_median	ci_lo	ci_hi	noninferior_0_01	share_gap_lt_minus001
classification	logreg	lac	24	0.901	0.001	-0.002	0.004	True	0.125
classification	logreg	aps	24	0.899	-0.001	-0.003	0.002	True	0.042
classification	rf	lac	24	0.902	0.002	-0.001	0.006	True	0.000
classification	rf	aps	24	0.901	0.001	-0.002	0.004	True	0.042
classification	lgbm	lac	24	0.905	0.005	0.001	0.008	True	0.042
classification	lgbm	aps	24	0.901	0.001	-0.001	0.004	True	0.042
classification	knn	lac	24	0.933	0.033	0.030	0.041	True	0.000
classification	knn	aps	24	0.901	0.001	-0.001	0.007	True	0.000
classification	mlp	lac	24	0.902	0.002	-0.004	0.006	True	0.208
classification	mlp	aps	24	0.899	-0.001	-0.003	0.002	True	0.042
regression	logreg	absres	12	0.899	-0.001	-0.004	0.003	True	0.083
regression	rf	absres	12	0.898	-0.002	-0.004	0.001	True	0.000
regression	lgbm	absres	12	0.899	-0.001	-0.003	0.001	True	0.083
regression	knn	absres	12	0.902	0.002	-0.001	0.008	True	0.000
regression	mlp	absres	12	0.900	0.000	-0.003	0.002	True	0.083

task	model	score	n	g_cov_median	gap_median	ci_lo	ci_hi	noninferior_0_01	share_gap_lt_minus001
regression	cqr	cqr	12	0.900	0.000	-0.003	0.009	True	0.083

S5 Null control (CLEAN datasets), α = 0.1

task	model	score	n	scp_cov_median	scp_cov_mean	max_abs_mcp_minus_scp
classification	logreg	lac	32	0.900	0.901	0.000
classification	logreg	aps	32	0.901	0.901	0.000
classification	rf	lac	32	0.902	0.902	0.000
classification	rf	aps	32	0.901	0.900	0.000
classification	lgbm	lac	32	0.900	0.900	0.000
classification	lgbm	aps	32	0.901	0.902	0.000
classification	knn	lac	32	0.941	0.943	0.000
classification	knn	aps	32	0.900	0.905	0.000
classification	mlp	lac	32	0.902	0.902	0.000
classification	mlp	aps	32	0.901	0.901	0.000
regression	logreg	absres	18	0.899	0.900	0.000
regression	rf	absres	18	0.902	0.903	0.000
regression	lgbm	absres	18	0.902	0.902	0.000
regression	knn	absres	18	0.901	0.901	0.000
regression	mlp	absres	18	0.902	0.902	0.000
regression	cqr	cqr	18	0.901	0.901	0.000

S6 Friedman tests across learners (novel coverage, SCP-R, DUP)

family	n	chi2	p	median_logreg	median_rf	median_lgbm	median_knn	median_mlp
classification_lac	24	51.5667	0.0000	0.8954	0.8936	0.8936	0.9259	0.8956
classification_aps	24	16.5667	0.0023	0.8988	0.9052	0.8978	0.9009	0.9002
regression_absres	12	4.3333	0.3628	0.8978	0.8919	0.8938	0.8982	0.8946

S7 Exploratory exposure strata

stratum	score	n_pairs	n_datasets	gap_median	gap_p10	gap_min	share_lt_m002	share_lt_m005	mcp_gap_median	mcp_gap_min	mcp_share_lt_m002
pi<0.1	absres	25	5	-0.0000	-0.0047	-0.0093	0.0000	0.0000	0.0017	-0.0048	0.0000
pi<0.1	aps	60	12	0.0023	-0.0038	-0.0088	0.0000	0.0000	0.0033	-0.0127	0.0000
pi<0.1	cqr	5	5	0.0018	-0.0038	-0.0042	0.0000	0.0000	0.0062	-0.0037	0.0000
pi<0.1	lac	60	12	0.0012	-0.0101	-0.0153	0.0000	0.0000	0.0020	-0.0187	0.0000
0.1<=pi<0.3	absres	20	4	-0.0090	-0.0223	-0.0306	0.1500	0.0000	-0.0007	-0.0064	0.0000
0.1<=pi<0.3	aps	35	7	-0.0017	-0.0123	-0.0379	0.0286	0.0000	-0.0001	-0.0138	0.0000
0.1<=pi<0.3	cqr	4	4	-0.0153	-0.0216	-0.0225	0.2500	0.0000	-0.0011	-0.0019	0.0000
0.1<=pi<0.3	lac	35	7	-0.0053	-0.0276	-0.0444	0.2000	0.0000	0.0006	-0.0106	0.0000
pi>=0.3	absres	15	3	-0.0128	-0.1638	-0.1692	0.3333	0.3333	-0.0017	-0.0578	0.3333
pi>=0.3	aps	25	5	-0.0074	-0.0311	-0.0490	0.1600	0.0000	0.0064	-0.0194	0.0000
pi>=0.3	cqr	3	3	-0.0172	-0.0332	-0.0372	0.3333	0.0000	-0.0002	-0.0007	0.0000
pi>=0.3	lac	25	5	-0.0063	-0.0652	-0.0687	0.4400	0.3200	0.0117	-0.0070	0.0000

S8 Verification run on solar_flare (200 additional splits)

model	n_splits	scp_novel_mean	mcp_novel_mean	mcp_novel_pooled	mcp_mean_ci	mcp_per_split_sd
lgbm	200	0.7807	0.9086	0.9090	[0.8992, 0.9179]	0.0683
logreg	200	0.7746	0.9065	0.9076	[0.8963, 0.9165]	0.0754

S9 Duplicate injection: medians by rate

`gap`/`mcp_gap`: novel-record coverage − 0.9 for split / novelty-Mondrian CP; `limit_gap`: memorization limit −απ/(1−π).

family	rate	gap	mcp_gap	limit_gap	exposure	cov_seen
classification_knn	0.0000	0.0471	0.0471	-0.0000	0.0000	
classification_knn	0.1000	0.0297	0.0333	-0.0121	0.1081	0.9751
classification_knn	0.2500	0.0311	0.0458	-0.0317	0.2410	0.9713
classification_knn	0.5000	0.0175	0.0383	-0.0680	0.4046	0.9692
classification_knn	1.0000	0.0150	0.0366	-0.1484	0.5973	0.9658
classification_lgbm	0.0000	0.0000	0.0000	-0.0000	0.0000	
classification_lgbm	0.1000	-0.0128	-0.0004	-0.0121	0.1081	1.0000
classification_lgbm	0.2500	-0.0261	0.0009	-0.0317	0.2410	0.9996
classification_lgbm	0.5000	-0.0567	0.0002	-0.0680	0.4046	0.9953
classification_lgbm	1.0000	-0.0791	0.0023	-0.1484	0.5973	0.9610
classification_logreg	0.0000	-0.0006	-0.0006	-0.0000	0.0000	
classification_logreg	0.1000	-0.0025	0.0021	-0.0121	0.1081	0.9217
classification_logreg	0.2500	-0.0043	0.0011	-0.0317	0.2410	0.9112
classification_logreg	0.5000	-0.0054	0.0021	-0.0680	0.4046	0.9100
classification_logreg	1.0000	-0.0084	0.0007	-0.1484	0.5973	0.9049
classification_rf	0.0000	0.0013	0.0013	-0.0000	0.0000	
classification_rf	0.1000	-0.0117	-0.0002	-0.0121	0.1081	1.0000
classification_rf	0.2500	-0.0268	0.0016	-0.0317	0.2410	1.0000
classification_rf	0.5000	-0.0614	0.0030	-0.0680	0.4046	1.0000
classification_rf	1.0000	-0.1275	0.0035	-0.1484	0.5973	0.9994
regression_knn	0.0000	0.0026	0.0026	-0.0000	0.0000	

family	rate	gap	mcp_gap	limit_gap	exposure	cov_seen
regression_knn	0.1000	-0.0040	0.0002	-0.0121	0.1081	0.9314
regression_knn	0.2500	-0.0073	0.0016	-0.0316	0.2401	0.9383
regression_knn	0.5000	-0.0141	0.0030	-0.0673	0.4024	0.9326
regression_knn	1.0000	-0.0256	0.0034	-0.1484	0.5973	0.9208
regression_lgbm	0.0000	-0.0015	-0.0015	-0.0000	0.0000	
regression_lgbm	0.1000	-0.0078	0.0016	-0.0121	0.1081	0.9773
regression_lgbm	0.2500	-0.0237	-0.0009	-0.0316	0.2401	0.9759
regression_lgbm	0.5000	-0.0421	0.0016	-0.0673	0.4024	0.9546
regression_lgbm	1.0000	-0.0527	0.0045	-0.1484	0.5973	0.9378
regression_logreg	0.0000	0.0010	0.0010	-0.0000	0.0000	
regression_logreg	0.1000	0.0016	0.0017	-0.0121	0.1081	0.9084
regression_logreg	0.2500	0.0006	0.0024	-0.0316	0.2401	0.9049
regression_logreg	0.5000	0.0035	0.0044	-0.0673	0.4024	0.9002
regression_logreg	1.0000	-0.0007	0.0005	-0.1484	0.5973	0.9022
regression_rf	0.0000	0.0035	0.0035	-0.0000	0.0000	
regression_rf	0.1000	-0.0116	-0.0001	-0.0121	0.1081	0.9955
regression_rf	0.2500	-0.0286	0.0004	-0.0316	0.2401	0.9921
regression_rf	0.5000	-0.0535	0.0019	-0.0673	0.4024	0.9898
regression_rf	1.0000	-0.1179	0.0049	-0.1484	0.5973	0.9803

S10 Capacity: medians by level

`gap`/`mcp_gap`: novel-record coverage − 0.9 for split / novelty-Mondrian CP; `limit_gap`: memorization limit −απ/(1−π).

family	level	gap	mcp_gap	limit_gap	exposure	cov_seen
classification_knn	0	0.1000	0.1000	-0.0137	0.1207	1.0000
classification_knn	1	0.0241	0.0367	-0.0137	0.1207	0.9760
classification_knn	2	0.0001	0.0108	-0.0137	0.1207	0.9523
classification_knn	3	-0.0032	0.0046	-0.0137	0.1207	0.9388
classification_logreg	0	-0.0087	0.0010	-0.0137	0.1207	0.9493
classification_logreg	1	-0.0083	-0.0005	-0.0137	0.1207	0.9434
classification_logreg	2	-0.0030	-0.0004	-0.0137	0.1207	0.9340
classification_rf	0	-0.0072	0.0022	-0.0137	0.1207	0.9673
classification_rf	1	-0.0048	0.0002	-0.0137	0.1207	0.9555
classification_rf	2	-0.0083	0.0031	-0.0137	0.1207	0.9357
classification_rf	3	-0.0071	-0.0006	-0.0137	0.1207	0.9297
regression_knn	0	-0.0002	0.0050	-0.0243	0.1956	0.9792
regression_knn	1	-0.0033	0.0007	-0.0243	0.1956	0.9509
regression_knn	2	-0.0045	-0.0015	-0.0243	0.1956	0.9391
regression_knn	3	-0.0030	-0.0008	-0.0243	0.1956	0.9329
regression_logreg	0	-0.0051	0.0010	-0.0243	0.1956	0.9201
regression_logreg	1	-0.0065	0.0010	-0.0243	0.1956	0.9189
regression_logreg	2	-0.0073	-0.0001	-0.0243	0.1956	0.9133
regression_rf	0	-0.0132	-0.0015	-0.0243	0.1956	0.9723
regression_rf	1	-0.0114	-0.0016	-0.0243	0.1956	0.9628
regression_rf	2	-0.0028	-0.0007	-0.0243	0.1956	0.9311
regression_rf	3	-0.0061	-0.0008	-0.0243	0.1956	0.9208

S10b Normalized-residual baseline (regression DUP datasets, alpha = 0.1)

Mean over 10 splits. Score |y - yhat| / (sigma(x) + beta), sigma = mean distance to 10 nearest training rows.

data_id	name	model	method	cov_novel	cov_seen	cov_all	exposure	size_novel_norm	size_all_norm
44959	concrete_compressive_strength	lgbm	MCP	0.9188	0.9875	0.9214	0.0403	0.9513	inf
44959	concrete_compressive_strength	lgbm	SCP	0.9192	0.9607	0.9209	0.0403	0.9538	0.9437
44959	concrete_compressive_strength	rf	MCP	0.9020	0.9875	0.9053	0.0403	1.0618	inf
44959	concrete_compressive_strength	rf	SCP	0.9056	0.9607	0.9078	0.0403	1.0744	1.0631
44962	forest_fires	lgbm	MCP	0.9325	1.0000	0.9346	0.0298	2.3814	inf
44962	forest_fires	lgbm	SCP	0.9325	1.0000	0.9346	0.0298	2.3673	2.3615
44962	forest_fires	rf	MCP	0.9356	1.0000	0.9375	0.0298	1.9488	inf
44962	forest_fires	rf	SCP	0.9346	1.0000	0.9365	0.0298	1.9335	1.9278
44963	physiochemical_protein	lgbm	MCP	0.8983	0.9026	0.8985	0.0473	2.2672	2.2374
44963	physiochemical_protein	lgbm	SCP	0.8957	0.9527	0.8984	0.0473	2.2488	2.2337
44963	physiochemical_protein	rf	MCP	0.8988	0.9017	0.8990	0.0473	2.1199	2.0499
44963	physiochemical_protein	rf	SCP	0.8939	0.9894	0.8984	0.0473	2.0798	2.0658
44964	superconductivity	lgbm	MCP	0.9009	0.9003	0.9007	0.3335	1.6095	1.4871
44964	superconductivity	lgbm	SCP	0.9066	0.8876	0.9002	0.3335	1.6661	1.4936
44964	superconductivity	rf	MCP	0.9008	0.9024	0.9013	0.3335	1.4951	1.4265
44964	superconductivity	rf	SCP	0.9115	0.8788	0.9006	0.3335	1.6094	1.4426
44966	solar_flare	lgbm	MCP	0.8500	0.9058	0.8986	0.8535	3.8773	2.4263
44966	solar_flare	lgbm	SCP	0.8976	0.9015	0.9019	0.8535	4.5336	2.3932
44966	solar_flare	rf	MCP	0.8642	0.9023	0.8981	0.8535	4.3156	2.8043
44966	solar_flare	rf	SCP	0.9069	0.8909	0.8939	0.8535	5.2725	2.7854
44970	QSAR_fish_toxicity	lgbm	MCP	0.9074	0.9433	0.9133	0.1856	2.2286	2.0839
44970	QSAR_fish_toxicity	lgbm	SCP	0.8981	0.9610	0.9094	0.1856	2.1365	2.0188
44970	QSAR_fish_toxicity	rf	MCP	0.8968	0.9260	0.9011	0.1856	2.1146	1.9655

data_id	name	model	method	cov_novel	cov_seen	cov_all	exposure	size_novel_norm	size_all_norm
44970	QSAR_fish_toxicity	rf	SCP	0.8885	0.9580	0.9017	0.1856	2.0188	1.9084
44971	white_wine	lgbm	MCP	0.9031	0.8872	0.8991	0.2338	2.6822	2.3386
44971	white_wine	lgbm	SCP	0.8743	0.9895	0.9012	0.2338	2.4438	2.3782
44971	white_wine	rf	MCP	0.8967	0.8910	0.8952	0.2338	2.5892	2.1740
44971	white_wine	rf	SCP	0.8657	0.9992	0.8968	0.2338	2.3475	2.2845
44972	red_wine	lgbm	MCP	0.9080	0.8909	0.9050	0.1806	2.9832	2.5499
44972	red_wine	lgbm	SCP	0.8809	1.0000	0.9025	0.1806	2.7420	2.6989
44972	red_wine	rf	MCP	0.9026	0.9004	0.9022	0.1806	2.8203	2.4780
44972	red_wine	rf	SCP	0.8863	1.0000	0.9069	0.1806	2.6533	2.6117
44976	sarcos	lgbm	MCP	0.8995	0.9095	0.9006	0.1090	0.5241	0.5125
44976	sarcos	lgbm	SCP	0.8961	0.9395	0.9008	0.1090	0.5158	0.5122
44976	sarcos	rf	MCP	0.8993	0.8979	0.8992	0.1090	0.5470	0.5097
44976	sarcos	rf	SCP	0.8869	0.9931	0.8985	0.1090	0.5187	0.5151
44984	cps88wages	lgbm	MCP	0.8965	0.9017	0.9010	0.8527	3.0157	3.7609
44984	cps88wages	lgbm	SCP	0.9988	0.8835	0.9005	0.8527	19.9767	5.9962
44984	cps88wages	rf	MCP	0.8982	0.9019	0.9013	0.8527	3.4465	3.9701
44984	cps88wages	rf	SCP	0.9987	0.8843	0.9011	0.8527	20.9491	6.2874
44990	brazilian_houses	lgbm	MCP	0.9033	0.9022	0.9032	0.0829	0.6377	0.6131
44990	brazilian_houses	lgbm	SCP	0.9004	0.9353	0.9033	0.0829	0.6279	0.6117
44990	brazilian_houses	rf	MCP	0.8995	0.9001	0.8995	0.0829	0.6119	0.5870
44990	brazilian_houses	rf	SCP	0.8970	0.9265	0.8994	0.0829	0.6044	0.5885
44993	health_insurance	lgbm	MCP	0.9026	0.9021	0.9026	0.0970	3.0965	3.0631
44993	health_insurance	lgbm	SCP	0.9124	0.7915	0.9007	0.0970	3.2024	3.0830
44993	health_insurance	rf	MCP	0.9031	0.9024	0.9030	0.0970	3.2356	3.2470
44993	health_insurance	rf	SCP	0.9165	0.7597	0.9013	0.0970	3.4218	3.2943

S11 Computational cost (Experiment G)

task	data_id	name	n_rows	n_features	t_keys_s	t_audit_s	t_taxonomy_s	t_scp_threshold_s	t_mcp_threshold_s	t_lgbm_fit_predict_s
classification	1486	nomao	30000	118	0.1647	0.0073	0.0061	0.0000	0.0001	2.7280
classification	1590	adult	30000	14	0.0673	0.0056	0.0063	0.0000	0.0000	0.9187
classification	23517	numerai28.6	30000	21	0.0086	0.0077	0.0063	0.0000	0.0000	0.8931
classification	41027	jungle_chess_2pcs_raw_endgame_complete	30000	6	0.0029	0.0064	0.0061	0.0000	0.0000	1.7293
classification	40668	connect-4	30000	42	0.1556	0.0063	0.0067	0.0000	0.0000	2.7043
regression	44976	sarcos	30000	21	0.0117	0.0079	0.0060	0.0000	0.0001	1.3192
regression	44963	physiochemical_protein	30000	9	0.0040	0.0072	0.0063	0.0000	0.0001	0.6004
regression	44974	video_transcoding	30000	18	0.0199	0.0064	0.0058	0.0000	0.0000	0.7358
regression	44979	diamonds	30000	9	0.0179	0.0064	0.0056	0.0000	0.0000	0.6299
regression	44975	wave_energy	30000	48	0.0244	0.0074	0.0062	0.0000	0.0000	2.8234

Fig. S1

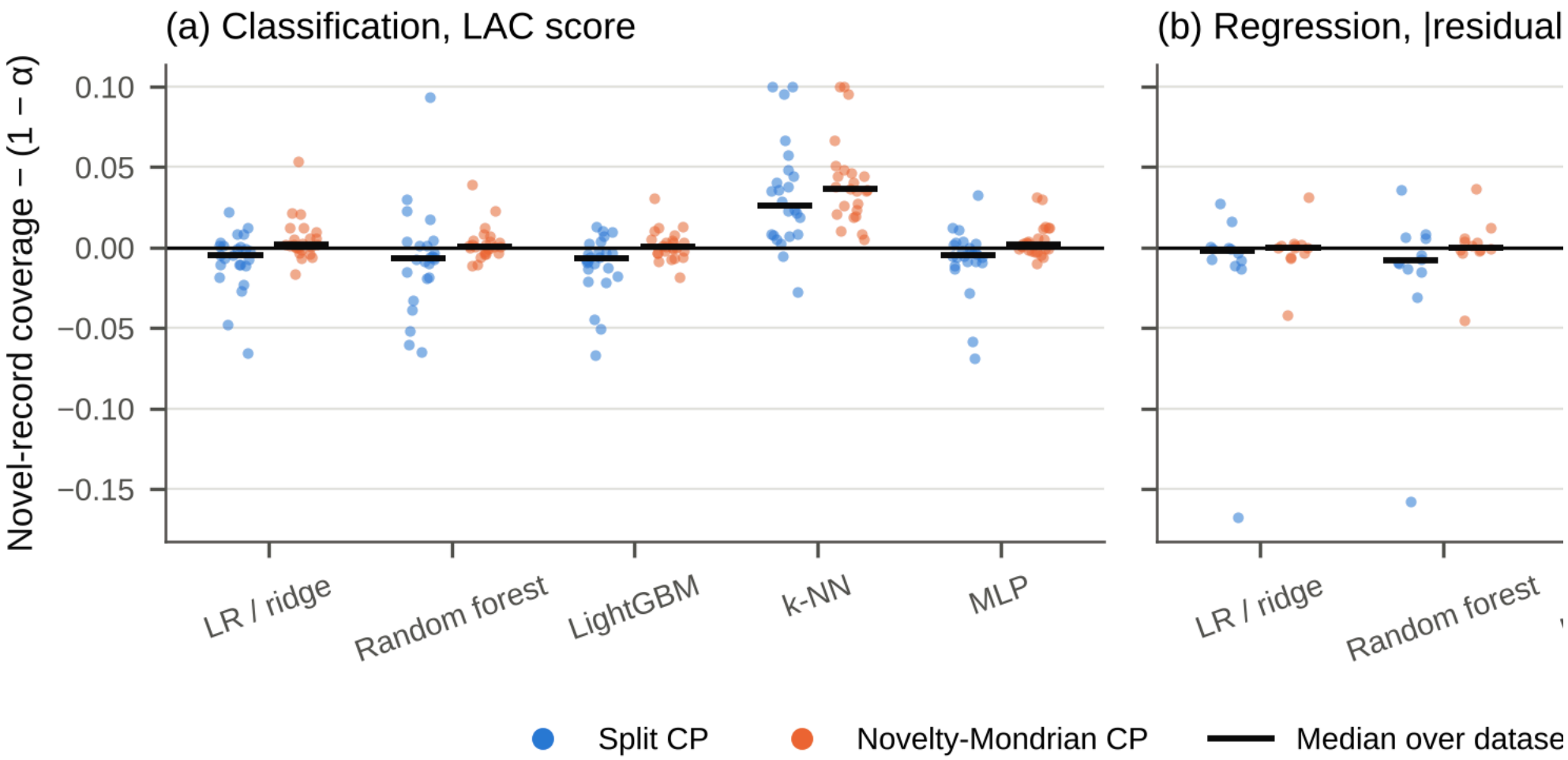


Fig. S1 Novel-record coverage gap per duplicated dataset (dots) and learner, split CP vs novelty-Mondrian CP, alpha = 0.1; bars: medians

S12 Computational environment

item	value
python	3.13.16 (main, Oct 1 2026, 15:40:24) [GCC 13.3.0]
platform	Linux-6.18.44-fc-v139-x86_64-with-glibc2.39
machine	x86_64
cpu_count	4
numpy	2.5.3
pandas	3.0.5
scipy	1.18.1
sklearn	1.9.1
lightgbm	4.7.0
time	2026-10-21 11:05:55